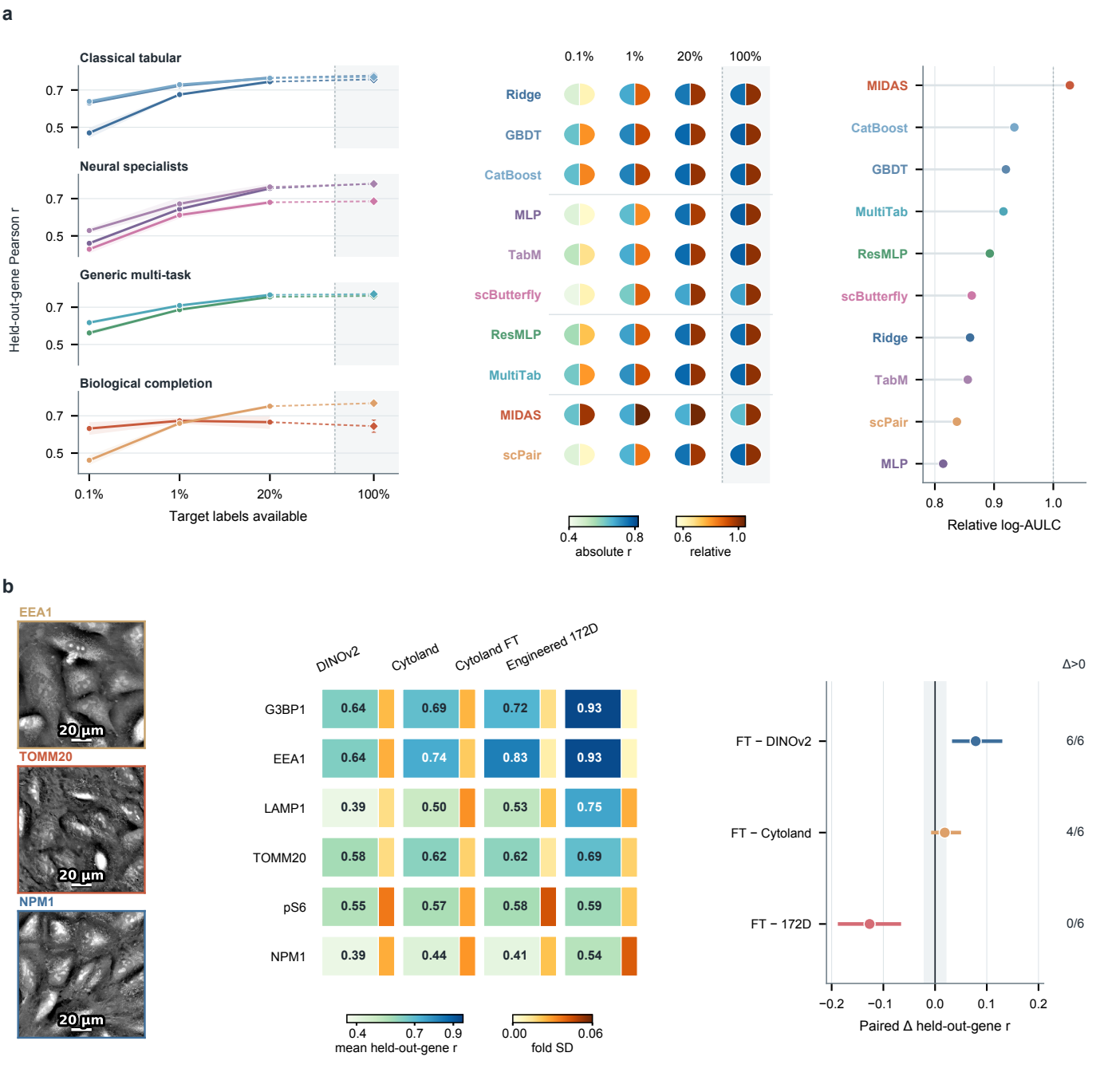

a
Classical tabular
Neural specialists
Generic multi-task
Biological completion
Held-out-gene Pearson r
0.7
0.5
0.1%
1%
20%
100%
Target labels available
0.1%
1%
20%
100%
Ridge
GBDT
CatBoost
MLP
TabM
scButterfly
ResMLP
MultiTab
MIDAS
scPair
0.4
0.8
absolute r
0.6
1.0
relative
MIDAS
CatBoost
GBDT
MultiTab
ResMLP
scButterfly
Ridge
TabM
scPair
MLP
0.8
0.9
1.0
Relative log-AULC
b
EEA1
TOMM20
NPM1
20 µm
DINOv2
Cytoland
Cytoland FT
Engineered 172D
G3BP1 0.64 0.69 0.72 0.93
EEA1 0.64 0.74 0.83 0.93
LAMP1 0.39 0.50 0.53 0.75
TOMM20 0.58 0.62 0.62 0.69
pS6 0.55 0.57 0.58 0.59
NPM1 0.39 0.44 0.41 0.54
0.4
0.7
0.9
mean held-out-gene r
0.00
0.06
fold SD
Δ>0
FT − DINOv2
FT − Cytoland
FT − 172D
6/6
4/6
0/6
−0.2
−0.1
0.0
0.1
0.2
Paired Δ held-out-gene r